

DMRG in era of MPS.

→ what is MPS?

→ what is DMRG?

→ how do they come together?

for N-body wavefunction, $\dim(\text{Hilbert space } H) = d^N$ ($d \rightarrow 2$)
 → not computationally efficient!

Assumption: low energy, low entanglement states of Hilbert space can be represented as a **MPS**

$$|\psi\rangle = \sum A^i \dots A^n |i \dots n\rangle$$

Why low entanglement? GS of local Hamiltonian is less entangled than random state. → AREA LAW

Local Hamiltonian $S_A \propto L^{d-1}$
 Random $S_A \propto L^d$ } surface vs volume scaling
 → only boundary of

Notations

c $c \rightarrow$ scalar

$$\frac{A}{i,j}$$

$A_{ij} \rightarrow$ value

each branch is another index.

v_i $v_i \rightarrow$ vector

$$\frac{A}{i,j}$$

$A_{ij}^c \rightarrow$ tensors...

Sample contraction calculations:

$$\sum_{j,k} A_{ijm} B_{jkn} = T_{ilm}$$

$\Rightarrow$



Complexity:

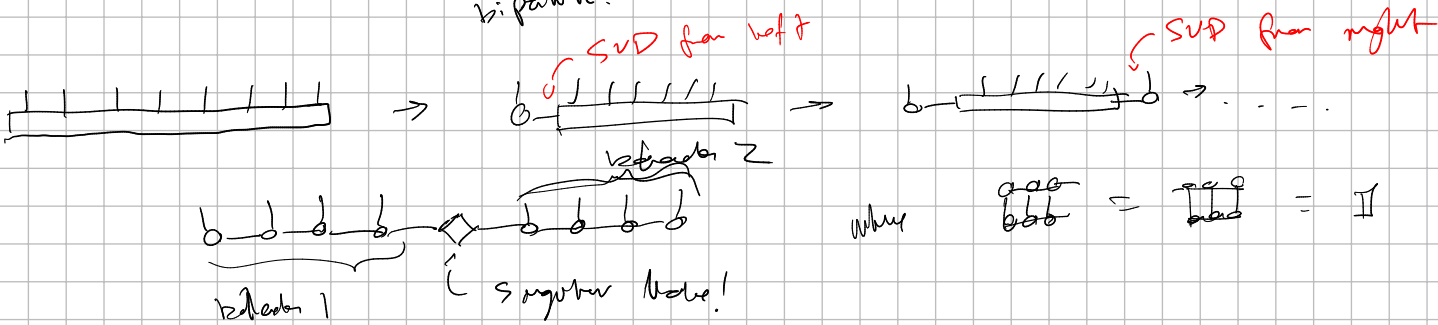
$\prod_{\text{conn.}} D_i \cdot \prod_{\text{open}} D_j \Rightarrow$ product of dimensionality of each "leg"

How do we get MPS from wavefunction? SVD!

→ Using SVD, keep decomposing until we get Schmidt decomposition.

$$|\psi\rangle = \sum \psi_{ij} |i\rangle |j\rangle \xrightarrow{\text{SVD}} \sum \lambda_i |i\rangle |j\rangle$$

is pairwise.



If wavefunctions become MPS, Hamiltonian become MPO

MPO:

$$W_{i,i+1}^{\sigma_i, \sigma_{i+1}} W_{i+1, i+2}^{\sigma_{i+1}, \sigma_{i+2}} W_{i+2, i+3}^{\sigma_{i+2}, \sigma_{i+3}} \dots \Rightarrow \begin{array}{c} \sigma' \\ | \\ i - \text{---} i+1 \\ | \\ \sigma \end{array} \begin{array}{c} \sigma' \\ | \\ i+1 - \text{---} i+2 \\ | \\ \sigma \end{array} \begin{array}{c} \sigma' \\ | \\ i+2 - \text{---} i+3 \\ | \\ \sigma \end{array}$$

We need to define behavior for 3 cases:

$$W_{\uparrow, \uparrow}^{\sigma, \sigma'} = \begin{array}{c} \sigma' \\ | \\ \uparrow - \text{---} \uparrow \\ | \\ \sigma \end{array}$$

$$W_{\uparrow, \downarrow}^{\sigma, \sigma'} = \begin{array}{c} \sigma' \\ | \\ \uparrow - \text{---} \downarrow \\ | \\ \sigma \end{array}$$

$$W_{\downarrow, \downarrow}^{\sigma, \sigma'} = \begin{array}{c} \sigma' \\ | \\ \downarrow - \text{---} \downarrow \\ | \\ \sigma \end{array}$$

We do this in the following manner: write all all possible terms of H with:

pairs on one site:

Block A	SU	Block B
1 — I —	σ^z	— I —
2 — σ^x —	σ^x	σ^x —
3 — X —	σ^x	— —
— —	—	— —
— —	—	— X —

e.g. $H = \sum \sigma_i^x \sigma_{i+1}^x + \sum \sigma_i^z$

all possible terms for each; to sum through.

these are all bases.

3 diff. unique entries for each: 3×3

$$\begin{array}{c} \text{SU} \\ \left. \begin{array}{c} - X - \\ - \frac{1}{2} \sigma^x - \\ - I - \end{array} \right\} \begin{array}{ccc} I & 0 & 0 \\ \sigma^x & 0 & 0 \\ \sigma^z & \frac{1}{2} \sigma^x & I \end{array} \left. \begin{array}{c} \\ \\ \end{array} \right\} \text{MPO!} \\ \text{--- I --- } \sigma^x \text{ --- } X \text{ --- } (R) \end{array}$$

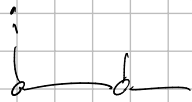
More Exercises:

S_1, S_2 :

$$H_{S_1, S_2} = \sum \frac{1}{2} (\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+) + \sigma_j^z \sigma_{j+1}^z$$

$$\left. \begin{array}{c} - I - \frac{1}{2} \sigma_j^+ \\ - \frac{1}{2} \sigma_j^+ \\ - I - \frac{1}{2} \sigma_j^- \\ - \frac{1}{2} \sigma_j^- \end{array} \right\} \begin{array}{c} \sigma_{j+1}^- \\ \sigma_{j+1}^- \\ \sigma_{j+1}^+ \\ \sigma_{j+1}^+ \end{array} \left. \begin{array}{c} \\ \\ \\ \end{array} \right\} \begin{array}{c} - I - \frac{1}{2} \sigma_j^z \\ - \frac{1}{2} \sigma_j^z \\ - I - \\ - X - \end{array} \left\{ \begin{array}{c} \sigma_{j+1}^z \\ \sigma_{j+1}^z \\ I \\ I \end{array} \right\} \begin{array}{c} \sigma_{j+1}^- \\ I \\ X \\ I \end{array}$$

What about boundary case?



$$w_I = \begin{pmatrix} \sigma^- \\ \sigma^+ \\ \sigma^z \end{pmatrix}$$



$$\sum (\sigma_i^+ \sigma_{i+1}^-) = \sum \frac{1}{2} (\sigma_i^+ \sigma_i^- + \sigma_i^- \sigma_i^+) + \sum \sigma_i^z \sigma_{i+1}^z$$

$$\left. \begin{array}{c|c} \sigma_i^- & \sigma_i^+ \\ \sigma_i^+ & \sigma_i^- \\ \sigma_i^z & \sigma_i^z \\ 0 & 1 \\ 1 & -1 \end{array} \right\}$$

$$\begin{pmatrix} \sigma_i^- \\ \sigma_i^+ \\ \sigma_i^z \\ 1 \\ 0 \end{pmatrix} w_I$$

$$\begin{pmatrix} \sigma_i^+ \\ \sigma_i^- \\ \sigma_i^z \\ 0 \\ 1 \end{pmatrix} w_I$$

From paper:

$$\hat{H} = S_1 \sum_i S_i^z S_{i+1}^z + S_2 \sum_i S_i^z S_{i+2}^z$$

S_1 :

$$\begin{array}{c|c} & G^z \\ \hline & G^z \end{array}$$

S_2 :

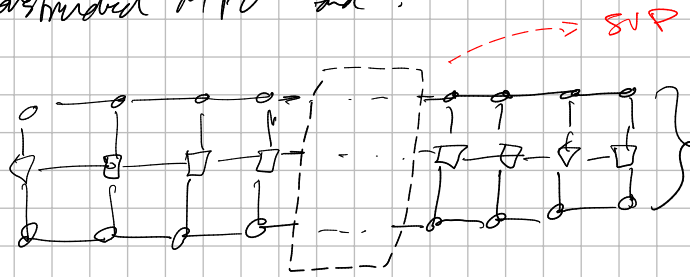
$$\begin{array}{c|c} G^z & I \\ \hline G^z I & G^z \\ \hline & G^z I G^z \end{array}$$

$$\begin{array}{c|c} \lambda & I \\ \hline & I \end{array}$$

$$\begin{array}{ccccc} G^z & I & 0 & G^z & 0 \\ G^z I & 0 & 0 & G^z & 0 \\ I & G^z & G^z & 0 & I \\ \hline 0 & 0 & 0 & I & 0 \\ \hline G^z & I G^z & I & -\lambda & \end{array}$$

DMRG

Take our constructed MPO and:



diagonalize, SVD,
optimize

We then sweep this, going as:



Why sweep twice? normalize from left + right!